6 Months Training TR-104 Full Stack Development

January, 2026

Day 4 – Component-Based Architecture and Parent-Child Communication

1. Objective of the Day

The objective was to understand Angular's component-based architecture and implement structured communication between components using @Input and @Output decorators.

2. Tasks Performed

- Studied the importance of dividing applications into reusable components
- Created a standalone Header component using Angular CLI
- Integrated the Header component into the main application
- Implemented parent-to-child communication using @Input decorator
- Passed data from parent component to child component
- Implemented child-to-parent communication using @Output decorator
- Used EventEmitter to emit events from child component
- Handled emitted events in the parent component
- Updated parent state based on child interaction

3. Tools / Technologies Used

- Angular Framework
- TypeScript
- HTML

4. Key Learnings

- Angular applications are built using reusable components
- Parent components control and manage application data
- @Input is used to pass data from parent to child
- @Output is used for communication from child to parent
- EventEmitter enables controlled event-based communication
- Angular follows unidirectional data flow

5. Challenges Faced

- Understanding the correct flow of data between components
- Implementing @Output and EventEmitter correctly

6. Solutions

- Practiced component interaction with multiple examples
- Followed structured approach to data flow implementation

7. Conclusion

- Successfully implemented parent-child communication in Angular
- Developed a clear understanding of component-based architecture